Tayyab Bilal

Senior AI Engineer

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Summary

Senior AI Engineer with 5 years of experience building applied AI and large language model (LLM) systems in Python. I build generative AI and agentic AI: autonomous planner/executor agents, ReAct tool-calling agents and multi-agent orchestration with LangGraph, LangChain and the Model Context Protocol (MCP), made safe with guardrails, human-in-the-loop (HITL) approvals and prompt-injection defence. I also build retrieval-augmented generation (RAG), LLM evaluation and LLMOps (cost tracking, CI/CD, end-to-end test harnesses) on FastAPI, Celery, Redis, PostgreSQL, Docker, AWS Bedrock, Azure OpenAI and GCP, plus machine learning (ML): fine-tuning (LoRA, QLoRA), NLP and computer vision in PyTorch.

Key Highlights

- **Top contributor of 11 engineers** on a multi-team AI platform, writing about 30% of its codebase.
- Designed a **deny-by-default, fail-closed human-in-the-loop** layer gating all 31 write tools of an autonomous agent.
- Built **LLM evaluation for AI agents**: an adversarial suite of 114 tests that closed 7 prompt-injection holes, plus end-to-end harnesses that check agent runs recover when a running service fails.
- Found that **only 68% of LLM provider spend was being billed** and built one cost tracker for every AI call.
- Built a **production conversational AI platform** at TransData (AWS Bedrock, Azure OpenAI), 99.5% uptime.

Experience

Senior AI Engineer Apr 2026 – Present

SWSAM Solutions | Lahore, Pakistan

AI-powered workspace platform for a Middle Eastern client

I build the AI service in Python and FastAPI (agents, chat, documents, workflows).

- Led the technical architecture for a few engineers, handing parts of the build to team members and guiding and reviewing their work.
- Built an autonomous planner/executor agent in LangGraph that turns a user goal into a durable, scheduled plan; code, not the LLM, assigns ids and schedules.
- Designed human-in-the-loop guardrails: every write action resolves per tenant to autonomous, suggest-only or disabled, through one shared check on all 31 write tools.
- Made agent runs safe to retry and run in parallel with PostgreSQL locks and a concurrent-run cap, plus a 90-turn budget and a loop guard; failed runs resume where they stopped.
- Hardened agents against prompt injection: event triggers pass ids only, and an adversarial suite of 114 tests closed 7 injection holes.
- Built end-to-end agent test harnesses on a dev tenant that fail a running service to check runs recover, passing 41/41 after fixes; wrote about half of all test code.
- Caught the planner silently running on the wrong default model in all 37 QA plans and moved it to Claude Sonnet, about 77% cheaper per plan.
- Verified the planner after the switch with a 24/24 benchmark and 20/20 end-to-end runs.
- Built a LangGraph ReAct discovery agent and used prompt engineering to make its parallel tools fire in one step, cutting cross-entity lookups from over 30 to 16 seconds or less.
- Chose grep-first agentic retrieval over vector RAG for workspace records (fuzzy matching plus PostgreSQL trigram search), with hybrid keyword and semantic search over Qdrant for documents.
- Measured 11,515 LLM call logs, found only 68% of provider spend was billed, and built one cost tracker and credit ledger with a CI check that fails any untracked LLM call.
- Built CI/CD from scratch in GitHub Actions, running tests in a network-isolated Docker image; change-aware caching cut PR setup from 514 to 61 seconds.
- For reliability, bounded every LLM call with concurrency caps, whole-call deadlines and stream caps, cutting the worst-case wait from 24 to 6 minutes.
- Mentored teammates through code reviews: reviewed 40 PRs with 409 inline comments, catching a cross-tenant security bug and a permanent model-lockout bug.

Senior Machine Learning Engineer Nov 2024 – Apr 2026

TransData | Lahore, Pakistan

Conversational AI, customer profiling and automated data reporting systems for enterprise clients

- Re-architected a monolithic agentic AI proof of concept into microservices, cutting deploys from over 2 hours to under 15 minutes.

- Built a production generative AI chat platform on AWS Bedrock agents and Azure OpenAI with pgvector memory and token-aware context trimming, at 99.5% uptime.
- Built LangGraph state-machine orchestration with LangChain tool calling for multi-step data ingestion and reporting, with error recovery and retries.
- Designed reusable MCP connectors for Slack, Box, Google Drive and SharePoint, cutting integrations from days to hours.
- Implemented agent-to-agent protocols for multi-agent tasks, plus n8n workflows with 20+ custom LangChain tools.
- Built hybrid RAG (dense + sparse retrieval, reranking, confidence scoring) on FAISS and Milvus, with sub-second search at millions of embeddings.
- Fine-tuned Qwen 7B with QLoRA for customer profiling, reaching over 95% accuracy against human profiles.
- Served the quantized model on AWS GPUs with dynamic batching, cutting inference cost 80% and raising throughput 5x.
- Benchmarked K-Means, DBSCAN and HDBSCAN, then shipped a two-stage clustering pipeline with drift detection and retraining at 89.1% cluster purity.
- Built voice agents on Deepgram, ElevenLabs and Azure AI Speech with provider fallback and sub-500 ms latency.

Machine Learning Engineer Feb 2022 – Sep 2024
Quid Sol | Lahore, Pakistan
Machine learning apps and chatbots for clients in healthcare, e-commerce, fintech and customer service

- Fine-tuned BERT, ResNet and YOLO models in PyTorch with transfer learning and hyperparameter tuning.
- Built invoice and contract extraction with fine-tuned MMOCR optical character recognition (OCR) and a BERT NER model, reaching 94% end-to-end accuracy.
- Cut document processing from 5 minutes to 30 seconds per file, routing low-confidence fields to human review.
- Built YOLO object detection trained on client data for inventory, quality control and security.
- Built image similarity search on ResNet embeddings and FAISS for visual recommendations and duplicate detection.
- Built a text + image recommender for a freelance marketplace; an online A/B test raised hire rate 12%.
- Built multi-platform chatbots (Messenger, WhatsApp, Slack, web) with Rasa NLU at under 200 ms latency.
- Built async FastAPI microservices for model serving and data ingestion, on PostgreSQL, MySQL and MongoDB.

Projects

Public implementations (open-source, on GitHub): autonomous-agent, agent-guardrails, grep-first-agent, llm-cost-meter, rag-quality-monitor, agent-kit, workflow-doctor, async-doc-ingestion, model-price-registry, fastapi-prod-hardening

Skills

LLMs & generative AI: AI agents, agentic AI, multi-agent systems, tool calling, ReAct, prompt engineering, structured outputs, guardrails, human-in-the-loop, LLM evaluation, OpenAI, Anthropic Claude, Google Gemini, AWS Bedrock, Azure OpenAI
Frameworks: LangChain, LangGraph, Model Context Protocol (MCP), n8n, PyTorch
Retrieval: RAG, hybrid search, BM25, reranking, embeddings, Qdrant, FAISS, Milvus, pgvector
Machine learning: fine-tuning (LoRA, QLoRA), quantization, NLP, NER, clustering, recommendation systems, computer vision (CV: YOLO, ResNet), OCR
Backend: Python, FastAPI, async, REST APIs, SSE, Celery, Redis, PostgreSQL, DynamoDB, MongoDB
Cloud & LLMOps: Docker, AWS (Bedrock, EC2, S3), Azure, GCP, CI/CD, GitHub Actions, pytest, LLM cost tracking, observability (Prometheus, LangSmith tracing), A/B testing, drift detection

Education

BSc Computer Science, University of Central Punjab, Lahore Oct 2017 – Aug 2021

Certifications

Anthropic (2026): Building with the Claude API, Claude with Amazon Bedrock, Model Context Protocol: Advanced Topics, Introduction to Model Context Protocol, Claude Code in Action, Introduction to subagents, Introduction to agent skills, AI Fluency: Framework & Foundations.

Technical Writing & Research

- Wrote 120+ design and architecture documents for an AI service, including “Graph RAG: A Case for Tool-Based Retrieval”, which shaped the retrieval layer.
- Designed the Three-Hook RAG evaluation framework (retrieval checks, sampled LLM-as-judge scoring, golden-set CI gate) and built it as an open-source project.